

ORIGINAL RESEARCH

Culture-Negative Infective Endocarditis in the Modern Diagnostic Era



Patient Characteristics and Heterogeneity in a Nationwide Cohort

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ABSTRACT

BACKGROUND Culture-negative infective endocarditis (CNIE) remains a diagnostic and therapeutic challenge. Current evidence on this disease is limited by small sample sizes, lack of granularity, and highly selected data.

OBJECTIVES This study sought to provide a granular characterization of CNIE in the modern diagnostic era from the first nationwide cohort of patients with infective endocarditis (IE).

METHODS This study used data from the nationwide Danish NIDUS (National Danish endocarditis stUdieS) registry, including all patients hospitalized with first-time left-sided IE (2016–2021). Patients were categorized as having CNIE or culture-positive infective endocarditis (CPIE) on the basis of blood cultures, valve tissue cultures, and polymerase chain reaction analyses. The study described patients' medical history, IE-specific disease features, clinical practice patterns, and treatment regimens.

RESULTS Of 2,875 IE patients, 212 (7.4%) had CNIE. Compared with CPIE, CNIE patients were more likely to have congenital heart disease (9.9% vs 2.9%) and less likely to have previous cancer (9.3% vs 14.7%). At admission, CNIE patients presented less frequently with sepsis (7.1% vs 24.6%) and fever (44.3% vs 61.7%), but they presented more often with embolism (25.9% vs 10.8%) and valvular insufficiency (11.3% vs 7.4%). The median size of vegetations was smaller (8 mm vs 10 mm) for patients with CNIE, whereas rates of prosthetic valve infection (22.2% vs 23.1%) and valve surgery (20.3% vs 21.8%) were similar. Positron emission tomography with computed tomography (PET-CT) was commonly used in both groups (67%–68%), and it was diagnostic in 13.1% of CNIE cases and 10.7% of CPIE cases. Among patients with prosthetic valve CNIE, the diagnostic yield of PET-CT was almost 50%.

CONCLUSIONS Using the first nationwide, unselected IE cohort, this study demonstrated a lower proportion of culture-negative cases than reported in previous studies. Patients with CNIE differed from CPIE in clinical presentation and disease characteristics, notably with less frequent fever and sepsis but higher embolic complication rates. However, culture-negative IE is highly heterogeneous and is better defined by distinct subgroups rather than by a single phenotype. (JACC. 2026;87:3060–3072) © 2026 by the American College of Cardiology Foundation.

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Patients with culture-negative infective endocarditis (CNIE) remain a long-standing clinical challenge.^{1,2} A positive blood culture result is a major diagnostic criterion for infective endocarditis (IE) and is often the initial indicator prompting clinicians to consider IE among the differential diagnoses.^{3,4} Furthermore, identifying the causative pathogen and determining its antimicrobial susceptibility are crucial for guiding optimal therapy; for example, patients with CNIE are typically not considered candidates for oral antibiotic regimens.⁴ However, in 8% to 30% of IE cases,^{5–10} blood culture results are negative, most commonly because of antibiotic exposure before blood culture collection, inadequate diagnostic testing, or infection with fastidious or nonculturable microorganisms.

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Two of the largest cohort studies investigating CNIE are the EURO-ENDO (European Society of Cardiology EuroObservational Research Programme on Endocarditis) and GAMES (Grupos de Apoyo al Manejo de la Endocarditis en España) studies,^{6,8} which report mortality and describe patient- and disease-related characteristics. However, together with several smaller cohorts,^{7,9,11} the existing epidemiologic literature on CNIE is limited by small sample sizes, inconsistent use of molecular diagnostics, variable access to advanced imaging, uneven case validation, and reliance on highly selected groups of patients from tertiary referral centers. Given the heterogeneity and clinical complexity of CNIE, robust epidemiologic characterization requires analysis within a large, unselected, and representative group of patients.

Therefore, using data from the newly established NIDUS (National Danish endocarditis studies) registry, we aimed to provide a modern characterization of the clinical features, treatment strategies, and distinct subgroups of patients with CNIE in a nationwide cohort of patients with IE.

METHODS

DATA SOURCES. The study cohort was drawn from the NIDUS database, a Danish nationwide retrospective registry that includes consecutive cases

classified as either definite IE or possible IE according to the modified Duke/European Society of Cardiology (ESC) 2015 diagnostic criteria.¹² The NIDUS database contains comprehensive information on patient demographics and baseline characteristics, echocardiographic findings, microbiological results, in-hospital treatment, site of infection, and clinical outcomes.¹³ All IE diagnoses were validated by a multidisciplinary team of health care professionals who systematically reviewed medical records of Danish patients who were assigned any of the following International Classification of Diseases-10th revision codes: DI33, DI38, or DI398. The database comprises records of 3,795 IE episodes and 3,557 individual IE patients who received an IE diagnosis in Denmark between January 1, 2016, and December 31, 2021. Among these patients, 79.6% met the criteria for definite IE and 20.4% met the criteria for possible IE. The cohort design has been described in detail previously.¹⁴

STUDY GROUP. We identified all episodes of IE between January 1, 2016, and December 31, 2021. Patients were categorized as those with CNIE and those with CPIE. CNIE was defined as IE with no bacterial pathogen identified with blood cultures, heart valve tissue cultures, or polymerase chain reaction (PCR). Patients were excluded if the infection involved valves other than the mitral or aortic valve, had a history of previous IE, or were aged <18 years, thus resulting in a cohort of patients with first-time left-sided IE. **Figure 1** shows the patient selection process.

Analyses were performed in a number of pre-specified subgroups of CNIE patients with the following: 1) abnormal native valves (ie, congenital heart disease or other native valve disease); 2) CNIE on prosthetic valves; 3) CNIE on native valves; 4) CNIE with embolic presentation at admission; 5) CNIE with a Duke definite diagnosis; and 6) CNIE with a Duke possible diagnosis.

FOLLOW-UP AND OUTCOMES. The objective was to describe the patient's medical history, IE-specific disease features, clinical practice patterns, and antibiotic treatment. Antibiotic treatment was defined as the antibiotic regimen administered on the first day

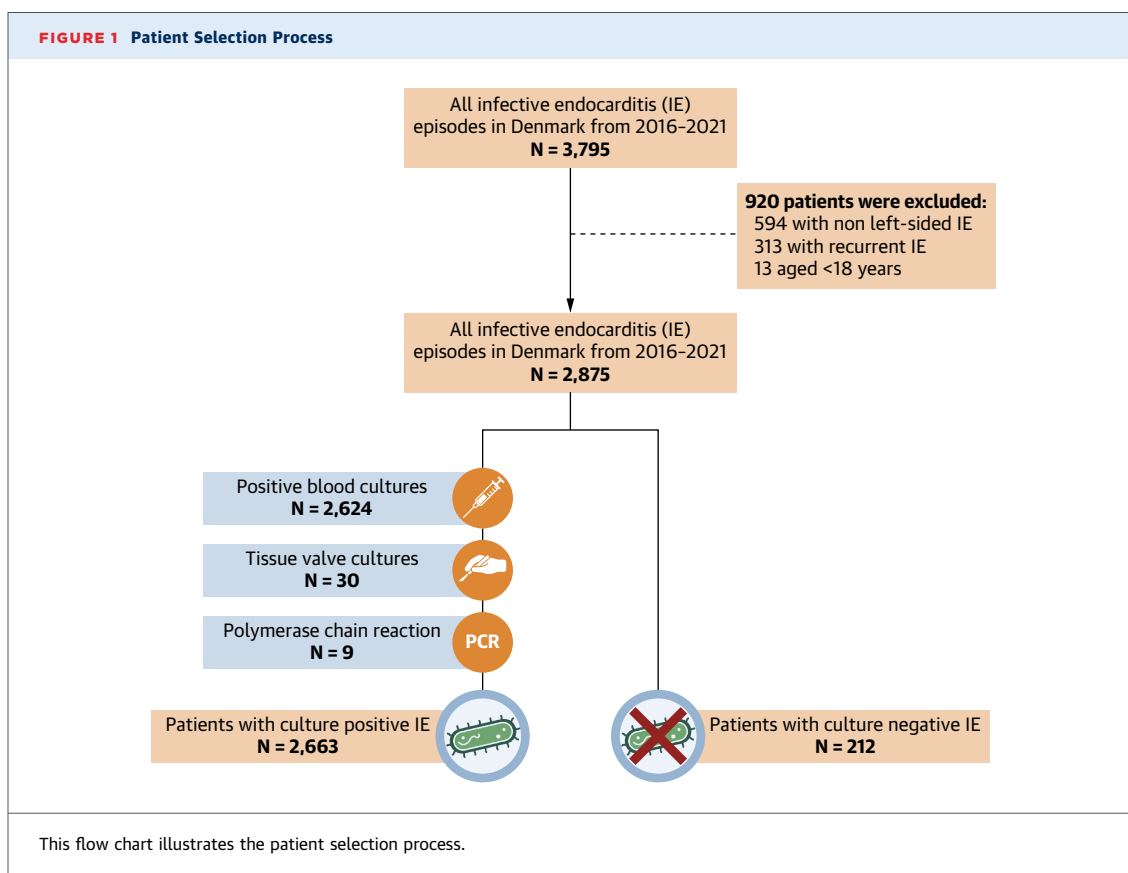
ABBREVIATIONS AND ACRONYMS

CNIE = culture-negative infective endocarditis

CPIE = culture-positive infective endocarditis IE, infective endocarditis

PCR = polymerase chain reaction

PET = positron emission tomography with computed tomography



of IE diagnosis, or, if antibiotic therapy at IE-specific doses was initiated before the day of diagnosis, the date of initiation of such treatment was used. Additionally, we aimed to examine in-hospital and 1-year all-cause mortality as well as the composite of the following events after discharge: 1) new or recurrent IE; 2) an embolic event; or 3) unplanned valvular heart surgery. For postdischarge outcomes, patients were followed up 12 months from the date of discharge. For the nonfatal outcomes, patients were censored at the date of mortality.

HANDLING OF MISSING INFORMATION. Missing data were present in <5% of all variables. Variables with >5% and >10% missing values are noted in **Tables 1 to 4**. Given their low proportion, missing values were not reported separately in these tables. For adjusted analyses, variables with missing data were categorized as 0, 1, or missing (ie, missing values were included as an explicit category) and were incorporated into the Cox proportional hazards models accordingly. This approach allowed all observations to be retained in the analysis without imputation.

STATISTICAL ANALYSIS. Baseline characteristics were compared between study groups. Categorical variables were reported as frequencies and percentages, with differences assessed using the chi-square test or the Fisher exact test when contingency tables contained sparse cells ($n < 5$). For categorical variables with >2 categories, only 1 P value was calculated. P values were not calculated when cell counts were <3. In these instances, exact numbers are not reported to protect patient identity. Continuous variables were presented as medians with quartile 1 to quartile 3 ranges (Q1-Q3), and differences were assessed using the Wilcoxon rank sum test. Crude in-hospital all-cause mortality was calculated as the number of patients who died before discharge divided by the total number of patients in the study group, and the difference between groups was calculated with the chi-square test. Adjusted in-hospital mortality rates were estimated using a cause-specific Cox proportional hazards model, controlling for age, sex, chronic kidney disease, diabetes mellitus, liver disease, chronic obstructive pulmonary disease, previous cancer diagnosis, active cancer, previous stroke, heart failure, prosthetic or

native valve IE, and valvular heart surgery during admission. Absolute mortality following discharge was estimated using the complement of the Kaplan-Meier survival function (1 - Kaplan-Meier). Absolute rates of the composite outcome following discharge were assessed with the Aalen Johansen estimator, accounting for the competing risk of death. Adjusted rates of outcomes following discharge were computed with a cause-specific Cox proportional hazards model, controlling for the same covariates as in the model for in-hospital mortality. The proportion hazards assumption was tested for all models, and no evidence of violation was observed ($P > 0.05$ for all models).

All statistical analyses were performed using SAS enterprise statistical software version 8.3 (SAS Institute) and RStudio version 2025.09.01+401 (RStudio). The level of statistical significance was recognized by a P value <0.05 .

SUPPLEMENTARY ANALYSES. To improve the robustness of our results, we performed 3 supplementary analyses. First, we extended the study group to include all left-sided IE cases (ie, also including recurrent IE cases). Second, we compared clinical presentation and diagnostic features between the CNIE group and the entire CPIE group, rather than the corresponding CPIE subgroups. Third, we examined 1-year mortality after stratifying the study group according to sex and surgical treatment or no surgical treatment.

ETHICS. The Danish National Ethics Committee approved the NIDUS registry (R-21025667), and the Danish Data Protection Agency (PACTIUS) approved data acquisition (P-2020-92).

RESULTS

PATIENTS' MEDICAL HISTORY. We identified 2,875 patients with first-time left-sided IE; 212 (7.4%) with CNIE (38.7% female, median age 72 years) and 2,663 (92.6%) with CPIE (33.1% female, median age 74 years).

Table 1 shows the patients' medical histories at baseline. Compared with patients with CPIE, patients with CNIE were slightly younger (72 years vs 74 years; $P < 0.01$) and with an overall comparable burden of comorbidities except a lower proportion with previous cancer (9.3% vs 14.7%; $P = 0.01$) and a higher proportion of congenital heart disease (9.9% vs 2.9%; $P < 0.01$). The most prevalent congenital heart diseases in the CNIE cohort were bicuspid aortic valve (67.5%) and ventricular septal defects (14.3%). CNIE patients were less often retired (73.8% vs 79.6%; $P = 0.04$) and more often

TABLE 1 Patients' Medical History

	CPIE (n = 2,663)	CNIE (n = 212)	P Value
Demographics			
Female	881 (33.1)	82 (38.7)	0.10
Median age, y	74.3 (65.6-81.1)	71.7 (61.1-79.6)	<0.01
Employment status			
Student	7 (0.3)	<3	0.04
Employed	395 (15.0)	44 (21.0)	
Unemployed	137 (5.2)	9 (4.3)	
Retired	2,097 (79.6)	155 (73.8)	
Medical history before admission			
Chronic kidney disease	448 (16.9)	31 (14.6)	0.61
Diabetes mellitus	624 (23.5)	38 (17.9)	0.16
Liver disease	137 (5.2)	4 (1.9)	0.09
Chronic obstructive pulmonary disease	191 (14.7)	35 (16.5)	0.72
Previous cancer	404 (15.2)	17 (8.0)	<0.01
Active cancer	247 (9.3)	16 (7.6)	0.43
Ischemic or hemorrhagic stroke	381 (14.3)	40 (19.0)	0.18
Chronic heart failure	413 (15.6)	28 (13.3)	0.60
Valvular heart disease	373 (14.1)	22 (10.4)	0.31
Congenital heart disease	77 (2.9)	21 (9.9)	<0.01
Bicuspid aortic valve	52 (67.5)	14 (66.7)	0.94
Ventricular septal defect	11 (14.3)	<3	N/A ^a
Truncus arteriosus	0	<3	N/A ^a
Unspecified left-sided heart valve anomaly	<3	<3	N/A ^a
Unspecified right-sided heart valve anomaly	0	<3	N/A ^a
Previous valvular heart surgery	714 (26.8)	56 (26.4)	0.90
Cardiac implantable electronic device	346 (13.1)	19 (9.0)	0.22
Coronary artery bypass graft	220 (8.3)	13 (6.1)	0.27
HIV or AIDS	8 (0.3)	0	0.67
Alcohol abuse	101 (3.8)	9 (4.3)	0.74
Active or previous intravenous drug abuse	63 (2.4)	3 (1.5)	0.10
Smoking status^b			
Active smoker	142 (18.6)	54 (25.5)	0.05
Previous smoker	958 (36.3)	74 (34.9)	
Never	1,071 (40.5)	79 (37.3)	
Self-reliant in activities of daily living	1,827 (68.9)	170 (80.2)	<0.01
Not self-reliant in activities of daily living	807 (30.3)	42 (19.8)	<0.01
Dependence on accessibility aid	251 (31.1)	12 (28.6)	
Domiciliary care	434 (53.8)	22 (52.4)	
Nursing home	118 (14.6)	8 (19.1)	
Unspecified need for aid	4 (0.5)	0	
Pharmacotherapy before admission			
Oral anticoagulant agents	394 (27.3)	28 (24.8)	0.77
Aspirin	483 (33.5)	52 (44.8)	0.04
Adenosine diphosphate inhibitors	233 (16.2)	21 (18.3)	0.79
Immunosuppressant agents	301 (11.5)	20 (9.6)	0.53

Values are n (%), median (Q1-Q3), or n. ^a P values were not calculated when cell counts were <3 . ^b $>5\%$ missing.

CNIE = culture-negative endocarditis; CPIE = culture-positive endocarditis; N/A = not applicable.

TABLE 2 Disease Characteristics

	CPIE (n = 2,663)	CNIE (n = 212)	P Value
Duke criteria			<0.01
Duke definite	2,279 (85.6)	48 (22.6)	
Duke possible	384 (14.4)	164 (77.4)	
Infective endocarditis type			0.75
Native valve	2,048 (76.9)	165 (77.8)	
Prosthetic valve	615 (23.1)	47 (22.2)	
Affected valve			0.51
Aortic valve	1,490 (56.0)	110 (51.9)	
Mitral valve	900 (33.8)	79 (37.3)	
Aortic and mitral valve	273 (10.3)	23 (10.9)	
Pathogen detection method			
Blood cultures	2,624 (95.5)	0	N/A
Median number of positive blood cultures	4 (3-4)	0	N/A
Surgically treated patients	581 (21.8)	43 (20.3)	0.51
Valve tissue culture performed	560 (96.4)	43 (100.9)	0.15
Positive valve tissue cultures	370 (66.1)	0	N/A
First-time pathogen detection with valve tissue culture	30 (5.4)	0	N/A
PCR performed	157 (27.3)	21 (48.8)	0.01
Positive PCR finding	107 (68.2)	0	N/A
First-time pathogen detection with PCR	9 (5.7)	0	N/A
Bacterial cause			
<i>Streptococcus</i> spp	953 (35.8)	0	N/A
<i>Staphylococcus aureus</i>	872 (32.8)	0	N/A
<i>Enterococcus</i> spp	482 (18.1)	0	N/A
<i>Coagulase-negative staphylococci</i>	159 (6.0)	0	N/A
<i>Aerococcus urinae</i>	24 (0.9)	0	N/A
<i>Escherichia coli</i>	15 (0.5)	0	N/A
<i>Aggregatibacter</i> spp	11 (0.4)	0	N/A
<i>Haemophilus influenzae</i>	10 (0.4)	0	N/A
<i>Cardiobacterium</i> spp	10 (0.4)	0	N/A
<i>Granulicatella</i> spp	10 (0.4)	0	N/A
<i>Klebsiella pneumoniae</i>	7 (0.3)	0	N/A
<i>Mycoplasma pneumoniae</i>	6 (0.2)	0	N/A
<i>Kingella kingae</i>	3 (0.1)	0	N/A
<i>Candida albicans</i>	<3	0	N/A
<i>Bartonella</i> spp	<3	0	N/A
Others (including fungi)	97 (3.6)	0	N/A
Median vegetation size, mm ^a	10 (6-15)	8 (6-12)	0.04
Osler nodes, petechiae, Janeway lesions, splinter hemorrhages, or retinal hemorrhages at admission ^a	206 (7.7)	17 (8.0)	0.88
Infection acquisition setting			0.03
Community acquired	1,832 (69.0)	133 (64.6)	
Hospital acquired	424 (16.0)	34 (16.5)	—
Health care associated (nonhospital)	169 (6.4)	9 (4.4)	—
Unknown	230 (8.7)	30 (14.6)	—
Recent dentist visit	115 (4.4)	11 (5.2)	0.31

Values are n (%), n, or median (Q1-Q3). ^a>10% missing.

PCR = polymerase chain reaction; spp = species; other abbreviations as in Table 1.

self-reliant in activities of daily living (80.2% vs 68.9%; $P < 0.01$).

PATHOGEN DETECTION METHOD AND USE OF POSITRON EMISSION TOMOGRAPHY WITH COMPUTED TOMOGRAPHY. The detected pathogen in patients with CPIE was identified for the first time by positive blood cultures in 2,624 (98.5%) patients, 30 (1.1%) by valve tissue cultures, and 9 (0.3%) by PCR (Figure 1).

Among the 43 (20.3%) patients with CNIE and 581 (21.8%) patients with CPIE who underwent surgery, valve tissue cultures were performed in 100% of CNIE cases and 96.4% of CPIE cases. Valve cultures identified or confirmed the causative pathogen in 66.1% of surgically treated CPIE patients, with first-time pathogen detection achieved in 30 patients. PCR analysis was performed in 48.8% of CNIE cases and in 27.3% of CPIE cases and resulted in first-time pathogen identification in 9 patients (Table 2).

Positron emission tomography with computed tomography (PET-CT) was performed equally often between the groups (68.4% vs 67.5%; $P = 0.86$) during admission and was the diagnostic modality in 10.7% of patients with CPIE and in 13.1% of patients with CNIE.

DISEASE CHARACTERISTICS. Table 2 summarizes disease characteristics. According to the modified Duke/ESC 2015 diagnostic criteria,¹² 22.6% of CNIE patients met the criteria for definite IE and 77.4% met the criteria for possible IE, compared with 85.6% and 14.4%, respectively, among CPIE patients ($P < 0.01$).

Compared with the CPIE group, the CNIE group displayed similar rates of native valve endocarditis (77.8% vs 76.9%) and prosthetic valve endocarditis (22.2% vs 23.1%; $P = 0.75$). Median vegetation size was smaller in CNIE patients (8 mm vs 10 mm; $P = 0.04$) and was equally often located on the aortic valve (56.0% vs 51.9%), the mitral valve (33.8% vs 37.8%), and both aortic and mitral valves combined (10.3% vs 10.9%). Infection was slightly less often community acquired in the CNIE group (64.6% vs 69.0%; $P = 0.03$) and more often of unknown source (14.6% vs 8.7%; $P = 0.03$).

CLINICAL PRESENTATION. Table 3 lists symptoms at presentation, treatment patterns, and IE-related complications. Patients had a similar median number of days from symptom onset to IE diagnosis (9 days vs 8 days; $P = 0.89$). At admission, CNIE patients less frequently presented with fever (44.3% vs 61.7%; $P < 0.01$) and sepsis (7.1% vs 24.6%; $P < 0.01$) (Figure 2). In contrast, patients with CNIE more often presented

with heart failure (4.7% vs 2.4%; $P = 0.05$) and valvular insufficiency (11.3% vs 7.4%; $P = 0.04$). Further, a high number of CNIE patients presented with embolisms (25.9% vs 10.8%; $P < 0.01$), most of which were embolisms to the central nervous system (77.1% for patients with CNIE and 68.2% for patients with C PIE).

HETEROGENEITY OF CNIE. Figure 3 summarizes key clinical and diagnostic features at presentation across CNIE subgroups compared with the corresponding C PIE subgroup. Across all subgroups, C PIE patients more frequently presented with fever and sepsis. In contrast, patients with CNIE consistently presented with more embolic events across all subgroups, with the highest frequencies observed among patients with abnormal valves and native valve IE.

PET-CT served as a common diagnostic modality among CNIE patients with prosthetic valve IE and those fulfilling Duke definite criteria, whereas no meaningful differences in PET-CT-based diagnosis were observed across the remaining subgroups. CNIE patients fulfilling Duke definite criteria exhibited more pronounced cardiac involvement, including higher rates of heart failure and valvular insufficiency, at presentation and were more likely to have congenital heart disease. In contrast, CNIE patients meeting only Duke possible criteria closely resembled C PIE patients, with the only principal differences being lower rates of fever and sepsis and higher rates of embolic events.

ANTIBIOTIC TREATMENT. Figure 4 summarizes the antibiotic treatment regimens administered on the first day of IE diagnosis in the CNIE group. On the day of the IE diagnosis, 60 (23.5%) patients initiated antibiotic monotherapy and 152 (76.5%) initiated antibiotic combination therapy. The 3 most common antibiotics in monotherapy were meropenem (9%), penicillin (6%), and cefuroxime (4%). The most common antibiotic combinations were gentamicin and meropenem (14%) and gentamicin and dicloxacillin and ampicillin (11%). The median duration of antibiotic treatment in relevant IE doses was lower among CNIE patients (34 days vs 39 days; $P = 0.03$) but similar numbers of patients were prescribed lifelong antibiotic prophylaxis following discharge (2.6% vs 3.7%; $P = 0.54$).

SURGICALLY AND NONSURGICALLY TREATED PATIENTS. Table 4 displays medical history, clinical presentation, and surgical details stratified into patients with and without valvular heart surgery. Heart valve surgery was performed in 16.8% of patients in the CNIE group and in 19.3% in the C PIE group. Overall, the surgically treated groups were significantly younger, carried a lower burden of

TABLE 3 Clinical Presentation, Treatment Patterns, and Complications

	CPIE (n = 2,663)	CNIE (n = 212)	P Value
Symptoms at admission			
Fever	1,584 (61.7)	90 (44.3)	<0.01
Myalgia ^a	690 (26.9)	55 (27.1)	0.92
Weight loss ^a	387 (15.1)	24 (11.8)	0.44
Dyspnea	834 (32.6)	69 (34.0)	0.60
Median symptom duration before IE diagnosis, d	8 (4-20)	9 (3-21)	0.89
Median time from admission to IE diagnosis, d	3 (2-6)	4 (1-8)	0.44
Complications at admission			
Heart failure	65 (2.4)	10 (4.7)	0.05
Conduction abnormalities	120 (4.5)	14 (6.6)	0.16
Valvular insufficiency	197 (7.4)	24 (11.3)	0.04
Sepsis	655 (24.6)	15 (7.1)	<0.01
Embolism	287 (10.8)	55 (25.9)	<0.01
CNS embolism	223 (68.2)	47 (77.1)	
Other embolism	64 (31.8)	8 (22.9)	
PET-CT was performed	1,797 (67.5)	145 (68.4)	0.86
PET-CT diagnosed IE	191 (10.7)	19 (13.1)	0.51
Complications during admission			
Acute kidney injury	474 (17.9)	36 (17.0)	0.84
Embolism	409 (15.4)	38 (17.9)	0.24
Heart failure	166 (6.2)	17 (8.2)	0.31
Conduction abnormalities	318 (11.9)	20 (9.4)	0.28
Severe or worsening valve insufficiency	262 (9.8)	19 (9.0)	0.40
Sepsis	301 (11.3)	19 (9.0)	0.30
Median duration of antibiotic treatment in IE doses, d	39 (28-45)	34 (27-43)	0.03
Lifelong prophylactic antibiotics after discharge	93 (3.7)	5 (2.8)	0.54
Median length of stay, d	35 (25-47)	40 (27-46)	0.40

Values are n (%) or median (Q1-Q3). ^a>5% missing.

CNS = central nervous system; IE = infective endocarditis; PET-CT = positron emission tomography with computed tomography; other abbreviations as in Table 1.

comorbidities, had larger vegetations, and presented with more advanced disease severity, including heart failure, valvular insufficiency, and embolism.

In the surgically treated groups, the most common indication for surgery was severe or progressive valvular insufficiency (55.8% vs 48.5%). In the CNIE group, the second most frequent indication was secondary embolic prophylaxis (25.3%). Notably, primary embolic prophylaxis was a significantly more common indication in the C PIE group than in the CNIE group (24.4% vs 9.3%). Patients in both groups could have more than 1 surgical indication.

Overall, patients with CNIE more frequently received mechanical prostheses (25.6% vs 17.7%) and less often received biological prostheses (46.5% vs 60.4%), although these differences did not reach statistical significance. Postoperative complications occurred in approximately one-half of the patients in both groups, with no notable differences.

TABLE 4 Characteristics of Surgically Treated Patients

	Surgically Treated Patients		Nonsurgically Treated Patients	
	CPIE (n = 581)	CNIE (n = 43)	CPIE (n = 2,082)	CNIE (n = 169)
Demographics				
Female	131 (22.6)	10 (23.3)	750 (36.0)	72 (42.6)
Median age, y	67.3 (57.0-73.9)	63.0 (53.1-72.3)	76.3 (68.3-82.6)	73.0 (63.9-80.9)
Medical history before admission				
Chronic kidney disease	52 (9.0)	4 (9.3)	396 (19.1)	27 (16.0)
Diabetes mellitus	111 (19.1)	3 (7.0)	513 (24.7)	35 (20.7)
Chronic obstructive pulmonary disease	64 (11.1)	7 (16.3)	327 (15.8)	28 (16.6)
Previous cancer	61 (10.5)	<3	343 (16.5)	16 (9.5)
Chronic heart failure	43 (7.4)	4 (9.3)	370 (17.8)	24 (14.3)
Valvular heart disease	68 (11.8)	8 (18.6)	305 (14.7)	14 (8.3)
Congenital heart disease	42 (7.2)	13 (30.2)	35 (1.7)	8 (4.7)
Previous valvular heart surgery	147 (25.3)	15 (34.9)	567 (27.2)	41 (24.3)
Cardiac implantable electronic device	30 (6.9)	3 (7.0)	306 (14.8)	16 (9.5)
Active smoking ^a	112 (19.3)	10 (23.3)	380 (18.4)	44 (26.0)
Self-reliance	530 (91.5)	41 (95.4)	1297 (62.6)	129 (76.3)
Clinical presentation at admission				
Fever	359 (67.4)	16 (41.0)	1225 (60.2)	74 (45.1)
Myalgia ^a	168 (31.6)	13 (33.3)	522 (25.7)	42 (25.6)
Weight loss ^a	124 (23.4)	6 (15.4)	263 (12.9)	18 (11.0)
Dyspnea	201 (39.9)	15 (38.5)	633 (31.2)	54 (32.9)
Heart failure	32 (5.5)	7 (16.3)	33 (1.6)	3 (1.8)
Conduction abnormalities	32 (7.2)	3 (7.0)	78 (3.8)	11 (6.5)
Valvular insufficiency	105 (18.1)	14 (32.6)	92 (4.4)	10 (5.9)
Sepsis	153 (26.3)	4 (9.3)	502 (24.1)	11 (6.5)
Embolism	105 (18.1)	15 (34.9)	182 (8.7)	40 (23.7)
Disease characteristics				
Duke criteria				
Duke definite IE	545 (93.8)	18 (41.9)	1734 (83.3)	30 (17.8)
Duke possible IE	36 (6.2)	25 (58.1)	348 (16.7)	139 (82.3)
IE type				
Native valve	445 (76.6)	29 (67.4)	1603 (77.0)	136 (80.5)
Prosthetic valve	136 (23.4)	14 (32.6)	479 (21.0)	33 (19.5)
Median vegetation size, mm ^b	13.0 (10.0-18.0)	12.5 (8.0-16.0)	9.0 (6.0-13.0)	8.0 (6.0-11.0)
Indications for surgery				
Severe or worsening valve insufficiency	282 (48.5)	24 (55.8)	N/A	N/A
Secondary emboli prophylaxis	97 (16.7)	11 (25.6)	N/A	N/A
Primary emboli prophylaxis	142 (24.4)	4 (9.3)	N/A	N/A
Vegetation size or mobility	123 (21.2)	9 (20.9)	N/A	N/A
Heart failure	53 (9.1)	7 (16.3)	N/A	N/A
Persistent bacteremia	31 (5.3)	<3 (-)	N/A	N/A
Myocardial abscess	115 (19.8)	5 (11.6)	N/A	N/A
Organism	5 (0.9)	0 (-)	N/A	N/A
Other	110 (18.9)	12 (27.9)	N/A	N/A
Surgical procedures				
Valvular repair	43 (7.4)	4 (9.3)	N/A	N/A
Mechanical prosthesis	103 (17.7)	11 (25.6)	N/A	N/A
Biological prosthesis	351 (60.4)	20 (46.5)	N/A	N/A
Homograft	9 (1.6)	0 (-)	N/A	N/A
Freestyle	99 (17.0)	9 (20.9)	N/A	N/A
Repair of fistula or ventricular septal rupture	33 (5.8)	3 (7.0)	N/A	N/A
Concomitant CABG	101 (17.5)	9 (20.9)	N/A	N/A

Continued on the next page

TABLE 4 Continued

	Surgically Treated Patients		Nonsurgically Treated Patients	
	CPIE (n = 581)	CNIE (n = 43)	CPIE (n = 2,082)	CNIE (n = 169)
Postoperative complications				
No complications	277 (47.7)	22 (51.2)	N/A	N/A
Stroke	6 (1.0)	<3 (-)	N/A	N/A
Dialysis treatment	64 (11.0)	<3	N/A	N/A
Delirium	90 (15.5)	5 (11.6)	N/A	N/A
Tracheostomy	25 (4.3)	4 (9.3)	N/A	N/A
Intubation >7	42 (7.2)	5 (11.6)	N/A	N/A
Reoperation for bleeding	106 (18.2)	8 (18.6)	N/A	N/A
Tamponade	36 (6.2)	3 (7.0)	N/A	N/A
Death	32 (5.5)	3 (7.0)	N/A	N/A

Values are n (%), median (Q1-Q3), or n. ^a>5% missing. ^b>10% missing.
CABG = coronary artery bypass graft; CNIE = culture-negative endocarditis; CPIE = culture-positive endocarditis; other abbreviations as in [Tables 1 and 3](#).

SECONDARY OUTCOMES. In-hospital mortality was 18.9% in patients with CNIE and 18.3% in patients with CPIE ($P = 0.84$). In adjusted analyses, this corresponded to an adjusted rate of mortality with an HR of 1.05 (95% CI: 0.74-1.48). At discharge, 2,347 patients were still alive: 172 (7.3%) with CNIE and 2,175 (92.7%) with CPIE. From the date of discharge, the 1-year all-cause mortality was 19.2% in patients with CNIE and 19.9% in patients with CPIE (log-rank $P = 0.85$) ([Figure 5](#)), corresponding to an adjusted HR of 1.07 (95% CI: 0.75-1.53).

[Supplemental Table 1](#) displays the absolute and adjusted 1-year risk of the composite of new or recurrent IE, embolic event, or unplanned valvular

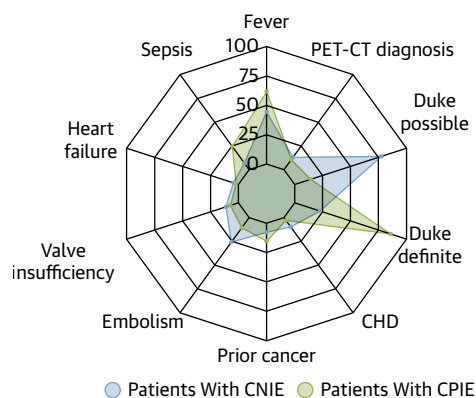
heart surgery. The rate of the composite outcome was 5.2% in patients with CNIE and 8.4% in patients with CPIE ($P = 0.23$), corresponding to an HR of 0.60 (95% CI: 0.30-1.19). When the composite outcome was analyzed by the individual components, patients with CNIE showed no significant difference in rates compared with those with CPIE.

SUPPLEMENTARY ANALYSES. Three supplementary analyses were performed. First, when extending the study group to include all patients with left-sided IE, including those admitted with recurrent IE, we identified 3,004 patients: 227 (7.6%) with CNIE and 2,777 (92.4%) with CPIE. [Supplemental Table 2](#) displays key clinical features between the groups. When comparing with the original cohort of only first-time IE cases, the comorbidity burden, clinical characteristics, and treatment patterns were comparable. Second, comparisons of clinical presentation and diagnostic features between CNIE subgroups and all CPIE patients, rather than the corresponding CPIE subgroup, yielded results similar to those observed when CNIE subgroups were compared with the corresponding CPIE subgroups ([Supplemental Figure 1](#)). Third, when examining the results according to sex and surgical treatment, the results demonstrated no difference in rates of mortality between CNIE and CPIE, although estimates were limited by low number and wide CIs ([Supplemental Table 3](#)).

DISCUSSION

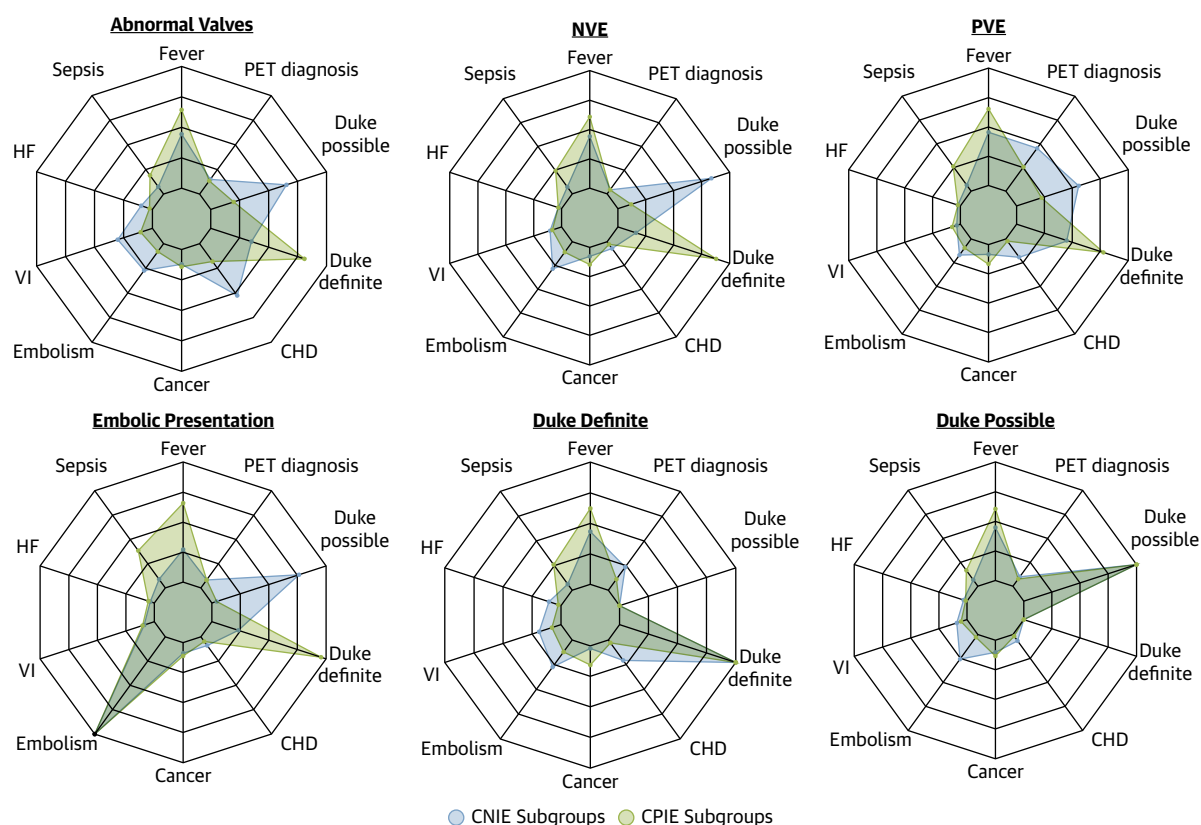
Drawing on the NIDUS registry, this study delivers the first nationwide, unselected assessment of left-sided CNIE, and it provides detailed insights into patient characteristics, disease features, CNIE subgroups, treatment patterns, and outcomes. Four

FIGURE 2 Clinical Presentation and Diagnostic Features



Comparison of the clinical presentation and diagnostic features patients with culture-negative infective endocarditis (CNIE) and culture-positive infective endocarditis (CPIE).
CHD = congenital heart disease; PET-CT = positron emission tomography with computed tomography.

FIGURE 3 Clinical Presentation and Diagnostic Features Among Subgroups



The clinical presentation and diagnostic features among the following subgroups of patients: 1) abnormal valves (ie, congenital heart disease [CHD] or other native valve disease); 2) native valve endocarditis (NVE); 3) prosthetic valve endocarditis (PVE); 4) patients presenting with emboli; 5) Duke definite infective endocarditis; and 6) Duke possible infective endocarditis. HF = heart failure; PET = positron emission tomography; VI = valve insufficiency; other abbreviations as in Figure 2.

main findings emerged. First, in this complete national cohort, approximately 1 in 13 patients with IE had CNIE. Second, the overall clinical presentation at admission differed between CNIE and CPIE, with CNIE patients exhibiting lower rates of fever and sepsis but higher frequencies of embolic events and valvular insufficiency. Third, subgroup analyses within CNIE revealed substantial disease heterogeneity, with marked differences in diagnostic yield and clinical presentation. Fourth, in-hospital mortality and 1-year mortality were comparable between CNIE and CPIE, and the risk of a composite outcome of recurrent or new IE, embolic events, and unplanned valvular heart surgery did not differ significantly between the groups.

PREVALENCE OF CNIE. The reported incidence of CNIE varies widely, ranging from approximately 8% to 30% in previous studies.⁵⁻¹⁰ The wide variation in reported CNIE rates is multifactorial and influenced

by selection of study groups, advances in diagnostic techniques, and the inclination to use empirical antibiotics before blood culture collection. Among studies reporting the lowest rates of CNIE are the Spanish GAMES cohort, which identified CNIE in 8.8% of 2,000 patients with IE,⁸ and the ICE (International Collaboration on Endocarditis) cohort, which reported CNIE in 11.1% of 2,781 patients.¹⁰ In the ESC-EURObservational Research Programme (EORP) EURO-ENDO international registry—the largest IE registry apart from NIDUS, CNIE was observed in 16.8% of 3,113 enrolled patients.⁶

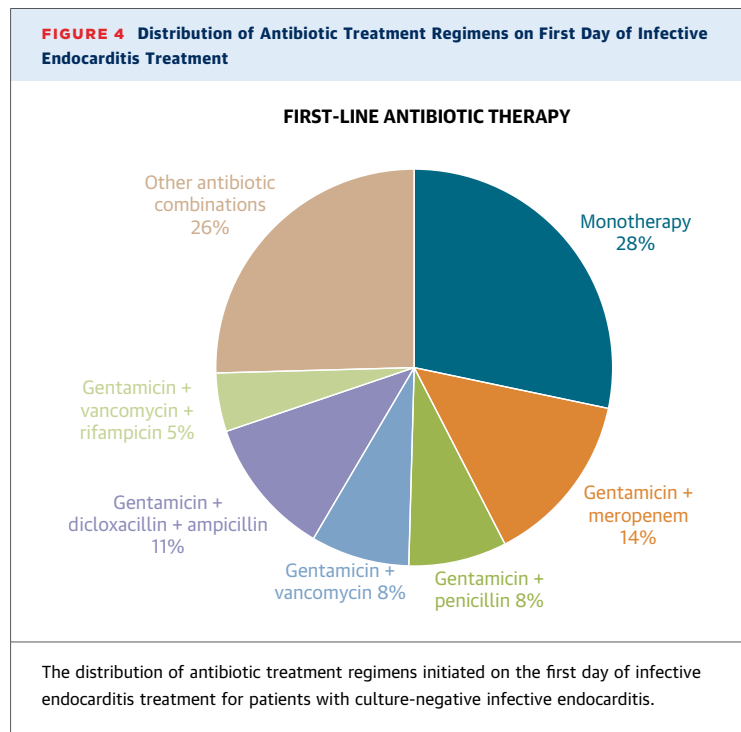
The EURO-ENDO registry included data from 3,113 patients across 156 centers in 40 countries between 2016 and 2019.⁶ It primarily represents tertiary care centers with voluntary participation and irregular inclusion rates across centers, thus introducing a significant selection bias, which is the main limitation of the registry. The highly selected data are reflected in the high proportion of surgical

interventions and the relatively low median age of patients in the cohort. In comparison, the NIDUS registry offers a more comprehensive and representative picture of the national IE group, thereby likely providing a more accurate estimate of the true epidemiology of both IE and CNIE. The NIDUS registry holds the largest unselected nationwide cohort to date (~3,500 IE patients enrolled between 2016 and 2021) with complete case validation.

Comparing CNIE prevalence, the present study, restricted to left-sided IE, identified a lower CNIE incidence of 7.4%. This lower incidence should be interpreted in light of 2 key factors. First, the present analysis was limited to left-sided IE to enhance comparability with existing literature because right-sided CNIE represents a distinct clinical entity with different patient characteristics and diagnostic considerations. Second, there is a generally high level of clinical awareness of IE in Denmark, and this results in a strong emphasis on obtaining comprehensive blood cultures before antibiotic initiation. Diagnostic evaluation follows a systematic approach, as illustrated in Figure 6. This approach entails obtaining blood cultures and performing transthoracic echocardiography at the time of clinical suspicion, followed by transesophageal echocardiography and PET-CT in all cases of IE to ensure accurate assessment of vegetations and to evaluate for extracardiac manifestations. Serologic testing is performed selectively, reflecting the extremely low prevalence of fastidious microorganisms such as *Coxiella burnetii* and *Bartonella* spp in Europe,¹⁵ but it is routinely applied when clinically indicated.

In patients who remain persistently culture negative and who are undergoing surgical treatment, pathogen identification may ultimately be achieved through valve tissue culture or PCR, which are both commonly applied, as discussed in the Results section. Thus, the CNIE patients in this study represent a thoroughly examined subgroup who had extensive efforts made on their behalf to detect the causative microorganism.

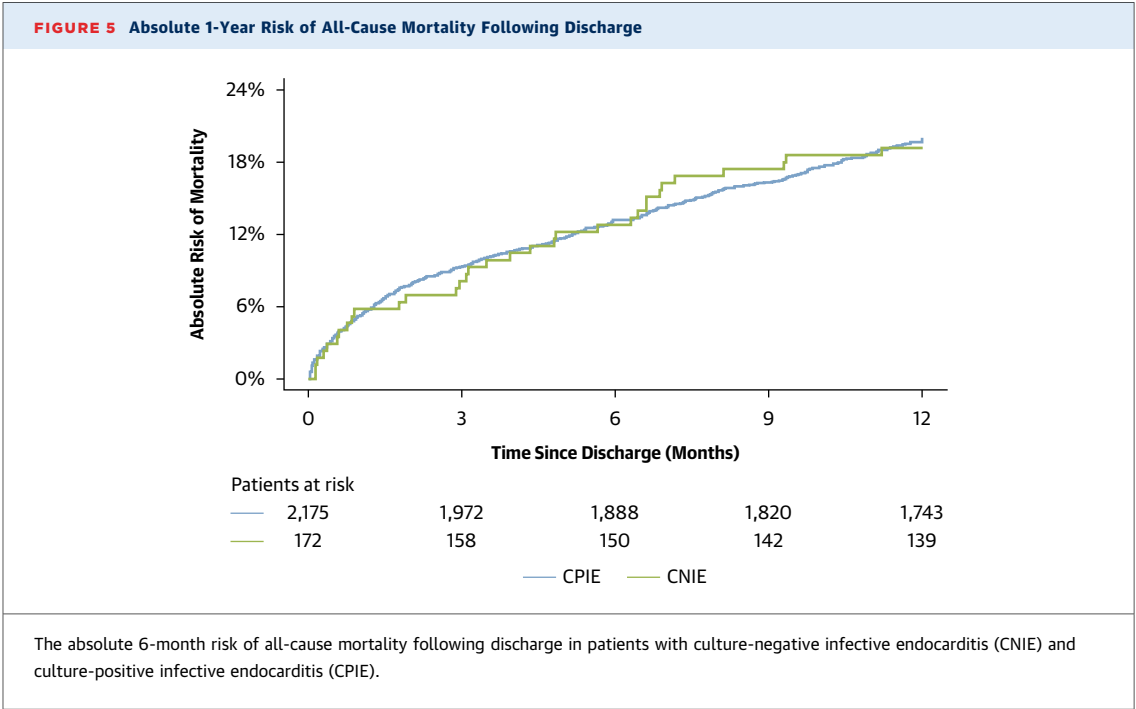
PATIENT CHARACTERISTICS AND HETEROGENEITY OF CNIE. Viewed as a whole, patients with CNIE were, on average, more functionally independent and had lower rates of previous malignant disease but a higher prevalence of congenital heart disease. Similar trends in patient characteristics were observed in both the EURO-ENDO and GAMES studies.^{6,8} At presentation, patients with CNIE were less likely to have fever or sepsis but more likely to exhibit advanced disease manifestations, including heart failure and valvular dysfunction. The former



presentation likely reflects, at least in part, the administration of empirical antibiotic therapy before establishment of the IE diagnosis, whereas the latter presentation may indicate a diagnostic delay in CNIE, with cardiac symptoms ultimately prompting recognition of the disease.

Notably, one of the most distinctive phenotypic differences between CNIE and CIE was the presence of embolic events at presentation in approximately 1 in 5 patients with CNIE, nearly twice the frequency observed in CIE. Consistent with this finding, secondary embolic prophylaxis constituted a surgical indication in 25% of CNIE patients undergoing valve surgery. Given the smaller median vegetation size and similar symptom duration in the CNIE cohort, the mechanism underlying the increased embolic presentation remains unclear. One possible interpretation is that CNIE follows a more indolent or subclinical course before diagnosis, in which embolic events, rather than classic infectious manifestations, serve as the initial trigger for clinical suspicion and subsequent diagnosis of IE. However, this explanation should be considered hypothesis generating and requires further investigation.

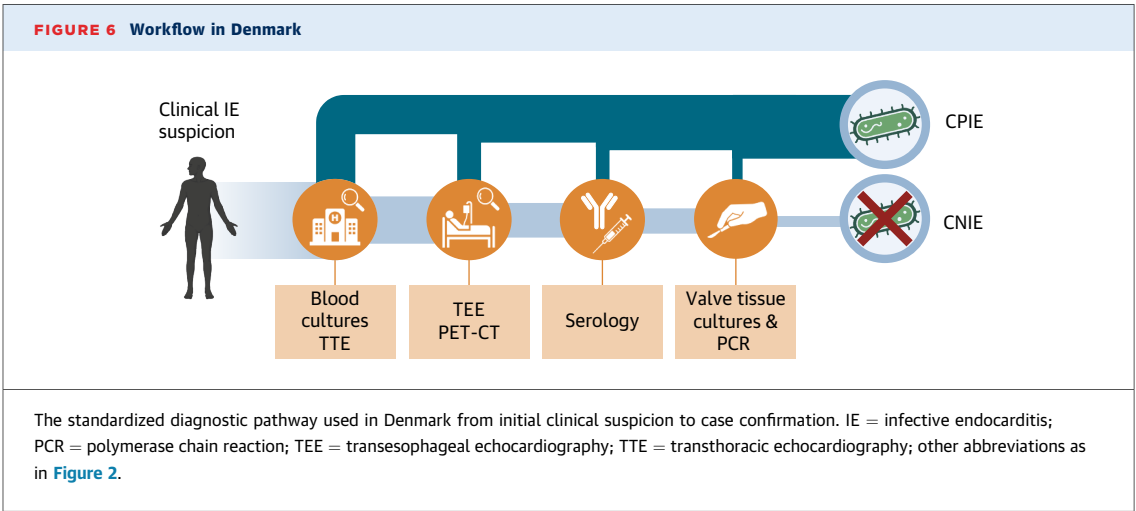
Other distinguishing features of the CNIE group were smaller vegetation size and more frequent unknown infection acquisition sites. Notably, left-sided CNIE and CIE patients had similar proportions of prosthetic valve IE, a finding that contrasts with



those of earlier studies, which reported higher prosthetic valve involvement in CNIE.

Subgroup analyses revealed heterogeneous CNIE phenotypes. Although certain features were consistent across all subgroups, most notably lower rates of fever and sepsis and a higher frequency of embolic events, other manifestations, including heart failure and valvular insufficiency, were predominantly observed in patients with preexisting valvular abnormalities and those meeting the Duke definite criteria. These findings reinforce the concept that CNIE is not a single disease entity but rather a heterogeneous and complex condition comprising multiple distinct entities. Leveraging the detailed and comprehensive data available in the NIDUS registry, this study provides a robust, real-world characterization of the epidemiology and clinical features of patients with CNIE.

SECONDARY OUTCOMES. CNIE has frequently been associated with worse outcomes than CPIE, commonly attributed to diagnostic delays, challenges in optimizing antimicrobial therapy, and delayed surgical evaluation.^{1,2,6-9,16} Large registry studies



such as GAMES and EURO-ENDO have reported higher short-term and midterm mortality among CNIE patients, largely explained by more advanced disease at diagnosis.^{6,8}

In our cohort, absolute and adjusted rates of in-hospital mortality and 1-year mortality were limited by a low number of events and were therefore underpowered. In the CNIE group, there were only ~40 in-hospital deaths and ~30 deaths within the first year following discharge, thus resulting in limited statistical precision and a reduced ability to detect clinically meaningful differences or assess equivalence. Accordingly, the comparable rates between groups should be interpreted cautiously and considered descriptive rather than inferential.

To our knowledge, only the EURO-ENDO study has previously assessed IE recurrence following discharge and reported similar 1-year rates of 2% to 4% in patients with CNIE and C PIE.⁶ In our study, examining survivors of the index admission, CNIE patients yielded a similar rate of the composite of new or recurrent IE, embolic events, and unplanned valvular heart surgery with a trend toward a lower associated rate. Although underpowered by low number of events, these findings include previously undescribed clinical characteristics and suggest that long-term outcomes in CNIE may be comparable to, or even better than, those observed in C PIE.

STUDY STRENGTHS. The primary strength of this study is the use of a nationwide cohort of patients with IE, including nearly all cases of definite or possible IE in Denmark from 2016 to 2021. Comprehensive clinical data allowed detailed assessment of baseline characteristics, diagnostic evaluation, clinical presentation, and treatment patterns. Validation through electronic medical record review minimized missing data and ensured complete follow-up.

STUDY LIMITATIONS. First, the observational design introduces the potential for selection bias and confounding by indication, and residual confounding may persist despite multivariable adjustment. Second, because the NIDUS registry is based on information extracted from clinical records, incomplete or missing documentation may have affected data quality; however, missingness was minimal for key variables. Importantly, information on antibiotic

exposure before hospital admission—an important determinant of blood culture yield and pathogen identification—was unavailable. In addition, although serologic and molecular diagnostic testing is performed in Denmark when clinically indicated, the registry records only pathogens identified through these methods and does not capture whether the tests were performed.

Furthermore, detailed intraoperative findings, including macroscopic valve appearance, histopathologic patterns of inflammation or tissue destruction, and concordance between surgical pathologic and echocardiographic findings, were not available in this retrospective cohort. In continuation, data on blood samples and echocardiographic information regarding vegetation morphology or timing of measurement relative to embolic events are not available. Finally, although this represents one of the largest IE cohorts to date, CNIE remains a relatively small subgroup in Denmark and Northern Europe. In particular, the limited number of deaths among CNIE patients renders mortality analyses underpowered, restricting the ability to detect meaningful differences or establish equivalence between groups and necessitating cautious, primarily descriptive interpretation of these results.

CONCLUSIONS

Using the first nationwide, unselected cohort of patients with IE, this study demonstrated a lower proportion of patients whose causative IE microorganism could not be identified. Among those patients who remained culture negative, we observed distinct medical histories, clinical presentations, and disease characteristics compared with C PIE. Most notably, patients with CNIE presented less frequently with fever and sepsis and had higher rates of embolic complications. Nevertheless, CNIE remains a highly heterogeneous condition and is more accurately characterized by distinct subgroups rather than by a single clinical phenotype.

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REFERENCES

1. Tunkel AR, Kaye D. Endocarditis with negative blood cultures. *N Engl J Med*. 1992;326(18):1215-1217.
2. Fosbøl EL. A double negative: culture-negative infective endocarditis. *Eur Heart J*. 2022;43(29):2781-2782.
3. Fowler VG, Durack DT, Selson-Suty C, et al. The 2023 Duke-International Society for Cardiovascular Infectious Diseases criteria for infective endocarditis: updating the modified Duke Criteria. *Clin Infect Dis*. 2023;77(4):518-526.
4. Delgado V, Ajmone Marsan N, de Waha S, et al. 2023 ESC guidelines for the management of endocarditis. *Eur Heart J*. 2023;44(39):3948-4042.
5. Ferrera C, Vilacosta I, Fernández C, et al. Reassessment of blood culture-negative endocarditis: its profile is similar to that of blood culture-positive endocarditis. *Rev Esp Cardiol (Engl Ed)*. 2012;65(10):891-900.
6. Kong WKF, Salsano A, Giacobbe DR, et al. Outcomes of culture-negative vs. culture-positive infective endocarditis: the ESC-EORP EURO-ENDO registry. *Eur Heart J*. 2022;43(29):2770-2780.
7. Murashita T, Sugiki H, Kamikubo Y, Yasuda K. Surgical results for active endocarditis with prosthetic valve replacement: impact of culture-negative endocarditis on early and late outcomes. *Eur J Cardiothorac Surg*. 2004;26(6):1104-1111.
8. Díez-Villanueva P, Muñoz P, Marín M, et al. Infective endocarditis: absence of microbiological diagnosis is an independent predictor of in-hospital mortality. *Int J Cardiol*. 2016;220:162-165.
9. Østergaard L, Voldstedlund M, Bruun NE, et al. Temporal changes, patient characteristics, and mortality, according to microbiological cause of infective endocarditis: a nationwide study. *J Am Heart Assoc*. 2022;11(16):e025801.
10. Murdoch DR, Corey GR, Hoen B, et al. International Collaboration on Endocarditis-Prospective Cohort Study Investigators. Clinical presentation, etiology, and outcome of infective endocarditis in the 21st century: the International Collaboration on Endocarditis-Prospective Cohort Study. *Arch Intern Med*. 2009;169(5):463-473.
11. Salsano A, Giacobbe DR, Del Puente F, et al. Culture-negative infective endocarditis (CNIE): impact on postoperative mortality. *Open Med (Wars)*. 2020;15(1):571-579.
12. Habib G, Lancellotti P, Antunes MJ, et al. ESC Scientific Document Group. 2015 ESC guidelines for the management of infective endocarditis: the Task Force for the Management of Infective Endocarditis of the European Society of Cardiology (ESC) endorsed by: European Association for Cardio-Thoracic Surgery (EACTS), the European Association of Nuclear Medicine (EANM). *Eur Heart J*. 2015;36(44):3075-3128.
13. Hadji-Turdeghal K, Graversen PL, Møller JE, et al. Patient characteristics, presentation, causal microorganisms, and overall mortality in the National Danish endocarditis stUdieS (NIDUS) registry. *Am Heart J*. 2025;285:119-132.
14. Graversen PL, Hadji-Turdeghal K, Møller JE, et al. National Danish endocarditis stUdieS - design and objectives of the NIDUS registry. *Am Heart J*. 2024;268:80-93.
15. Christodoulou M, Malli F, Tsaras K, Billinis C, Papagiannis D. A narrative review of Q fever in Europe. *Cureus*. 2023;15(4):e38031.
16. DeSimone DC, Garrigos ZE, Marx GE, et al. Blood culture-negative endocarditis: a scientific statement from the American Heart Association. *J Am Heart Assoc*. 2025;14(8):e040218.

KEY WORDS culture-negative infective endocarditis, epidemiology, infective endocarditis, outcomes, patient characteristics

APPENDIX For supplemental tables and a supplemental figure, please see the online version of this paper.